

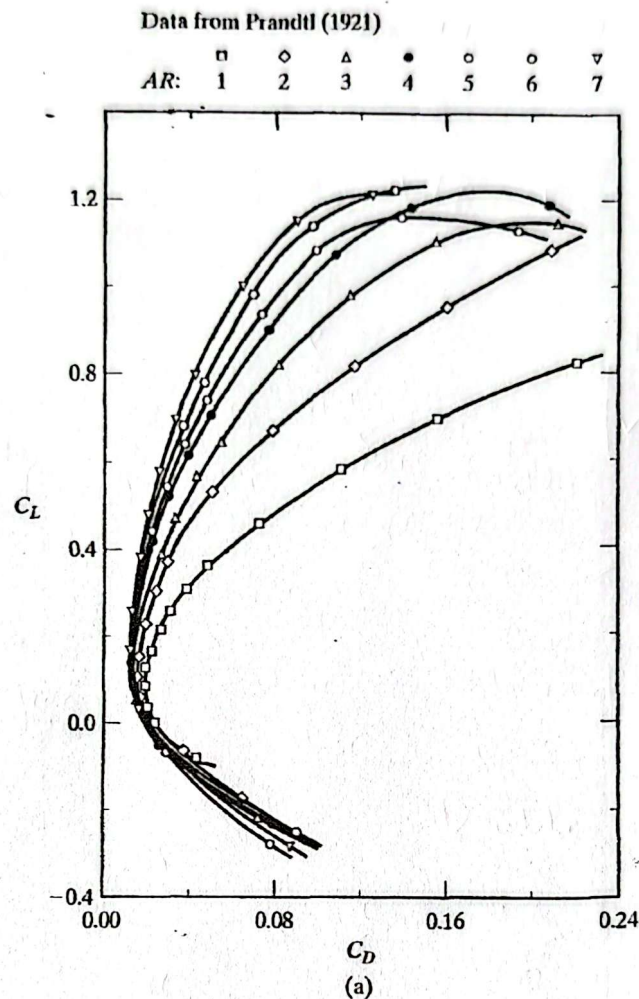


Figure 7.9 Effect of the aspect ratio on the drag polar for rectangular wings (AR from 1 to 7): (a) measured drag polars.

The results of converting the coefficient-of-lift measurements using equation (7.20) in terms of a wing whose aspect ratio is 5 (i.e., $AR_2 = 5$) are presented in Fig. 7.10b. Again, the converted curves collapse into a single correlation. Therefore, the validity of equation (7.20) is experimentally verified.

7.3.3 Technique for General Spanwise Circulation Distribution

Consider a spanwise circulation distribution that can be represented by a Fourier sine series consisting of N terms:

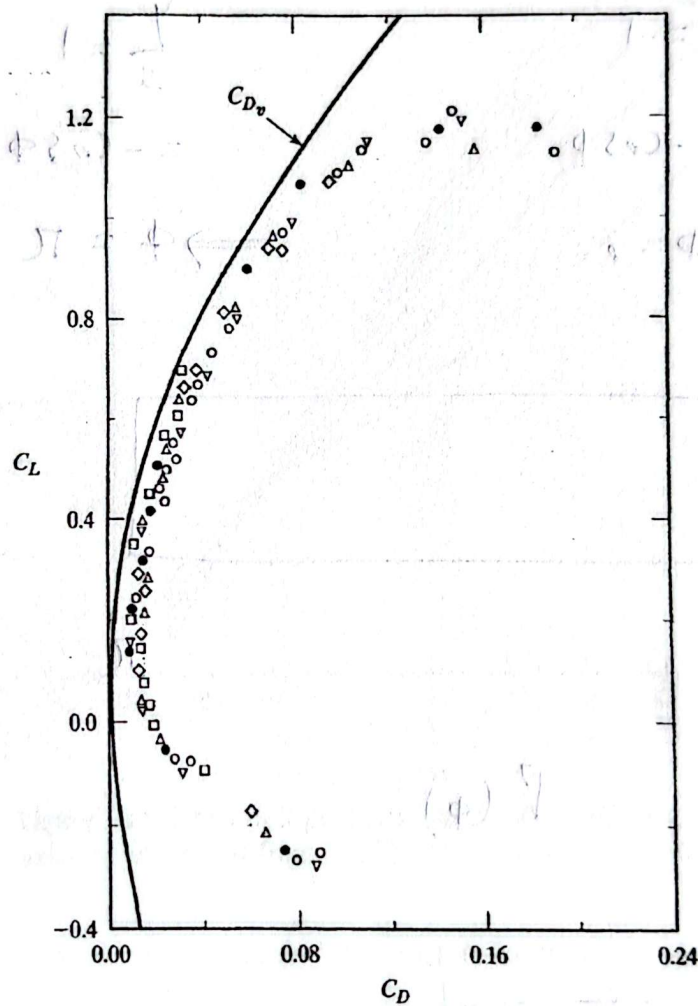
$$\Gamma(\phi) = 4sU_\infty \sum_{n=1}^N A_n \sin n\phi \quad (7.21)$$

$\underbrace{\hspace{1.5cm}}_{2b}$

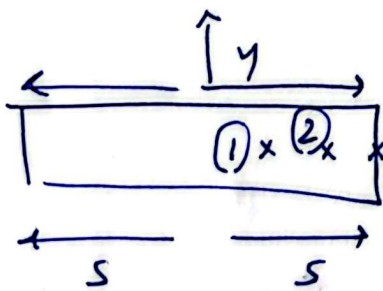
— Theoretical induced drag, eq. (7.16)

Data from Prandtl (1921)

AR:	1	2	3	4	5	6	7
	□	◇	△	●	○	○	▽



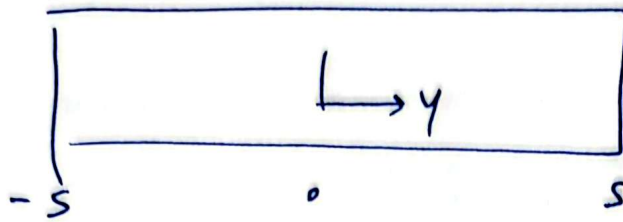
(b)

Figure 7.9 (continued) (b) drag polars converted to $AR = 5$.

As was done previously, the physical spanwise coordinate (y) has been replaced by the ϕ coordinate:

$$\frac{y}{s} = -\cos \phi$$

A sketch of one such Fourier series is presented in Fig. 7.11. Since the spanwise lift distribution represented by the circulation of Fig. 7.11 is symmetrical, only the odd terms remain.



$$\frac{y}{s} = -1$$

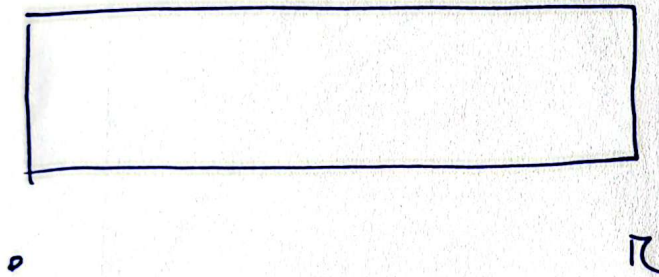
$$= -\cos \phi$$

$$\Rightarrow \phi = 0$$

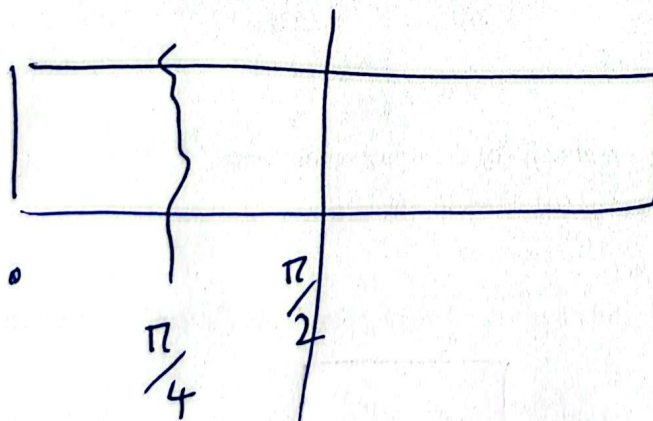
$$\frac{y}{s} = 1$$

$$= -\cos \phi$$

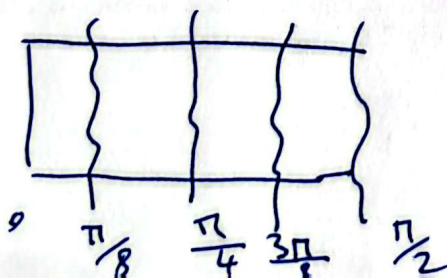
$$\Rightarrow \phi = \pi$$



$$\Gamma(\phi)$$



$$\begin{aligned} \Gamma &\propto \sum A_n \sin(n\phi) \\ &= A_1 \sin \phi \\ &\quad + A_3 \sin 3\phi \end{aligned}$$



$$\begin{aligned} \Gamma &\propto A_1 \sin \phi + A_3 \sin 3\phi \\ &\quad + A_5 \sin 5\phi + A_7 \sin 7\phi \end{aligned}$$

Data from Prandtl (1921)

AR: □ 1 ◇ 2 ▲ 3 ● 4 ○ 5 ○ 6 ▼ 7

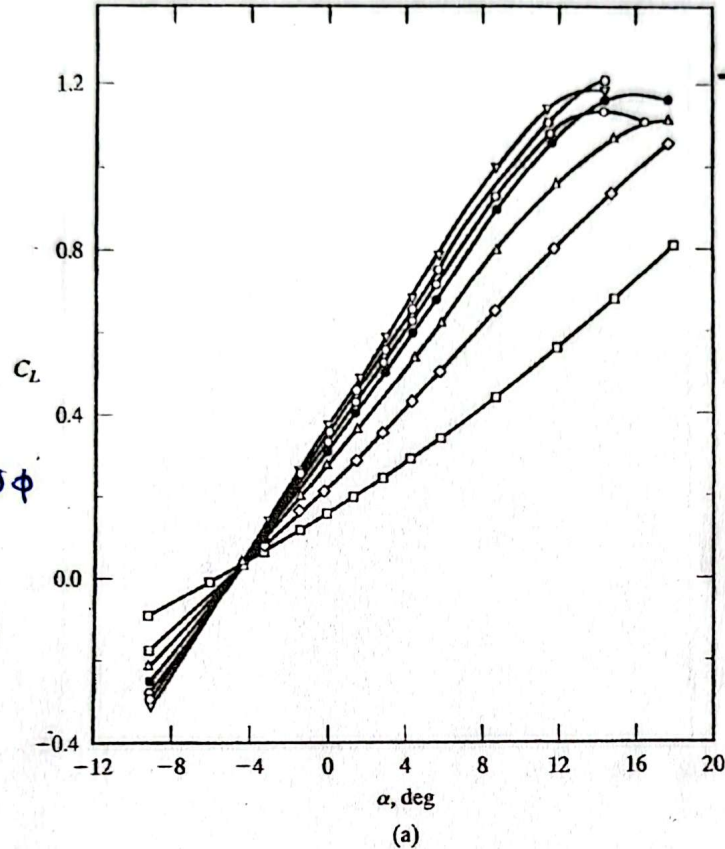


Figure 7.10 Effect of aspect ratio on the lift coefficient for rectangular wings (AR from 1 to 7): (a) measured lift coefficients.

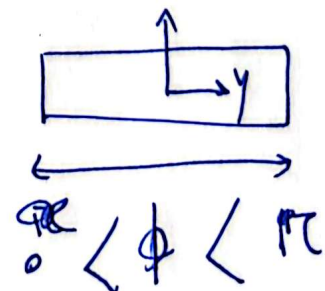
The section lift force [i.e., the lift acting on that spanwise section for which the circulation is $\Gamma(\phi)$] is given by

$$l(\phi) = \rho_{\infty} U_{\infty} \Gamma(\phi) = 4 \rho_{\infty} U_{\infty}^2 s \sum_{n=1}^N A_n \sin n\phi \quad (7.22)$$

To evaluate the coefficients $A_1, A_2, A_3, \dots, A_N$, it is necessary to determine the circulation at N spanwise locations. Once this is done, the N -resultant linear equations can be solved for the A_n coefficients. Typically, the series is truncated to a finite series and the coefficients in the finite series are evaluated by requiring the lifting-line equation to be satisfied at a number of spanwise locations equal to the number of terms in the series. This method, known as the collocation method, will be developed in this section.

Recall that the section lift coefficient is defined as

$$C_l(\phi) = \frac{\text{lift per unit span}}{\frac{1}{2} \rho_{\infty} U_{\infty}^2 c}$$



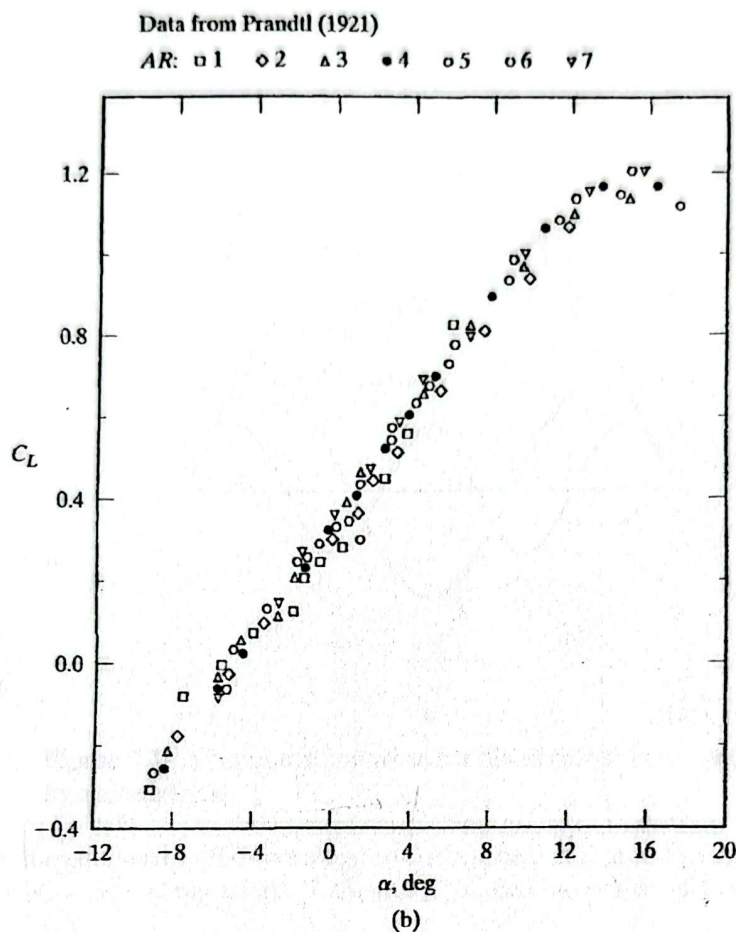


Figure 7.10 (continued) (b) Lift correlations converted to $AR = 5$.

Using the local circulation to determine the local lift per unit span, we obtain

$$C_l(\phi) = \frac{\rho_{\infty} U_{\infty} \Gamma(\phi)}{\frac{1}{2} \rho_{\infty} U_{\infty}^2 c} = \frac{2\Gamma(\phi)}{U_{\infty} c} \quad (7.23)$$

It is also possible to evaluate the section lift coefficient by using the linear correlation between the lift and the angle of attack for the equivalent two-dimensional flow. Thus, referring to Fig. 7.12 for the nomenclature, we have

$$C_l = \left(\frac{dC_l}{d\alpha} \right)_0 (\alpha_e - \alpha_w) \quad (7.24)$$

We now have two expressions for calculating the section lift coefficient at a particular spanwise location ϕ . We will set the expression in equation (7.23) equal to that in equation (7.24) to form equation (7.25).

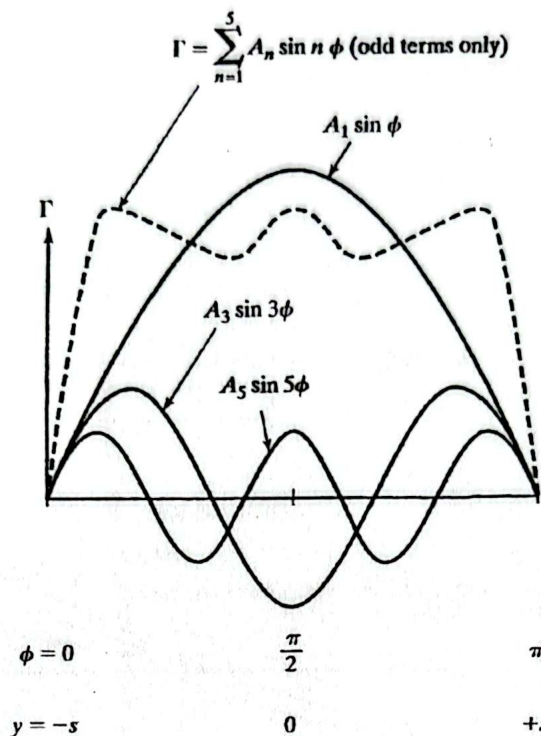


Figure 7.11 Symmetric spanwise lift distribution as represented by a sine series.

Let the equivalent lift-curve slope $(dC_l/d\alpha)_0$ be designated by the symbol a_0 . Note that since $\alpha_e = \alpha - \epsilon$, equations (7.23) and (7.24) can be combined to yield the relation

$$\frac{2\Gamma(\phi)}{c(\phi)a_0} = U_\infty[\alpha(\phi) - \alpha_{0l}(\phi)] - U_\infty\epsilon(\phi) \quad (7.25)$$

For the present analysis, five parameters in equation (7.25) may depend on the spanwise location ϕ (or, equivalently, y) at which we will evaluate the terms. The five parameters are (1) Γ , the local circulation; (2) ϵ , the downwash angle, which depends on the circulation distribution; (3) c , the chord length, which varies with ϕ for a tapered wing planform; (4) α , the local geometric angle of attack, which varies with ϕ when the wing is twisted (i.e., geometric twist, which is illustrated in Fig. 5.7), and (5) α_{0l} , the zero lift angle of attack, which varies with ϕ when the airfoil section varies in the spanwise direction (which is known as *aerodynamic twist*). Note that

$$U_\infty\epsilon = -w = -\frac{1}{4\pi} \int_{-s}^{+s} \frac{d\Gamma/dy}{y - y_1} dy$$

Using the Fourier series representation for Γ and the coordinate transformation, we obtain

$$-w = U_\infty \frac{\sum n A_n \sin n\phi}{\sin \phi}$$

Downwash for generalized lift dist.

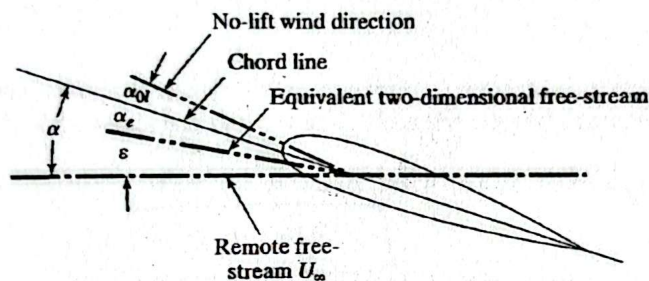
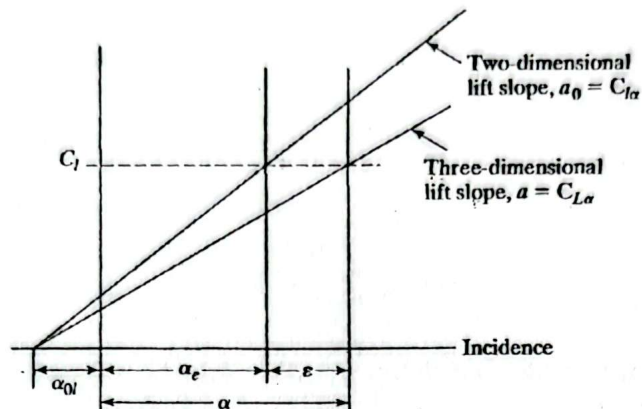
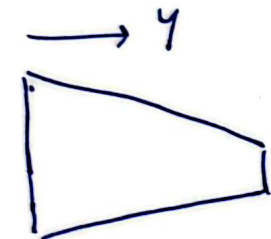


Figure 7.12 Nomenclature for wing/airfoil lift.



$$C = C(y)$$

Equation (7.25) can be rewritten as

$$\frac{2\Gamma}{ca_0} = U_\infty(\alpha - \alpha_{0l}) - U_\infty \frac{\sum n A_n \sin n\phi}{\sin \phi}$$

Since $\Gamma = 4sU_\infty \sum A_n \sin n\phi$, the equation becomes

chord length

$$\frac{8s}{ca_0} \sum A_n \sin n\phi = (\alpha - \alpha_{0l}) - \frac{\sum n A_n \sin n\phi}{\sin \phi}$$

$$\frac{y}{s} = -\cos \phi$$

Defining $\mu = ca_0/8s$, the resultant governing equation is

$$\mu = \frac{ca_0}{8s}$$

$$\mu(\alpha - \alpha_{0l}) \sin \phi = \sum A_n \sin n\phi (\mu n + \sin \phi) \quad (7.26)$$

which is known as the monoplane equation. If we consider only symmetrical loading distributions, only the odd terms of the series need be considered. That is, as shown in the sketch of Fig. 7.11,

$$a_0: c_\ell (\text{airfoil})$$

$$\Gamma(\phi) = 4sU_\infty (A_1 \sin \phi + A_3 \sin 3\phi + A_5 \sin 5\phi + \dots)$$

$$s = \frac{b}{2}$$

$$\left\{ \text{If the wing is rectangular} \right\}$$

$$\mu = \frac{ca_0}{8s}$$

7.3.4 Lift on the Wing

$$L = \int_{-s}^{+s} \rho_{\infty} U_{\infty} \Gamma(y) dy = \int_0^{\pi} \rho_{\infty} U_{\infty} s \Gamma(\phi) \sin \phi d\phi$$

Using the Fourier series for $\Gamma(\phi)$ gives us

$$L = 4\rho_{\infty} U_{\infty}^2 s^2 \int_0^{\pi} \sum A_n \sin n\phi \sin \phi d\phi$$

Noting that $\sin A \sin B = \frac{1}{2} \cos(A - B) - \frac{1}{2} \cos(A + B)$, the integration yields

$$L = 4\rho_{\infty} U_{\infty}^2 s^2 \left\{ A_1 \left[\frac{\phi}{2} + \frac{\sin 2\phi}{4} \right] \Big|_0^{\pi} + \sum_{n=3}^{\infty} \frac{1}{2} A_n \left[\frac{\sin(n-1)\phi}{n-1} - \frac{\sin(n+1)\phi}{n+1} \right] \Big|_0^{\pi} \right\}$$

The summation represented by the second term on the right-hand side of the equation is zero, since each of the terms is zero for $n \neq 1$. Thus, the integral expression for the lift becomes

$$L = (4s^2) \left(\frac{1}{2} \rho_{\infty} U_{\infty}^2 \right) A_1 \pi = C_L \left(\frac{1}{2} \rho_{\infty} U_{\infty}^2 \right) (S)$$

$$C_L = A_1 \pi \cdot AR$$

(7.27)

The lift depends only on the magnitude of the first coefficient, no matter how many terms may be present in the series describing the distribution.

7.3.5 Vortex-Induced Drag

$$\begin{aligned} D_v &= - \int_{-s}^{+s} \rho_{\infty} w \Gamma dy \\ &= \rho_{\infty} \int_0^{\pi} \underbrace{\frac{U_{\infty} \sum n A_n \sin n\phi}{\sin \phi}}_{-w} \underbrace{4s U_{\infty} \sum A_n \sin n\phi}_{\Gamma} \underbrace{s \sin \phi}_{dy} d\phi \\ &= 4\rho_{\infty} s^2 U_{\infty}^2 \int_0^{\pi} \sum n A_n \sin n\phi \sum A_n \sin n\phi d\phi \end{aligned}$$

The integral

$$\int_0^{\pi} \sum n A_n \sin n\phi \sum A_n \sin n\phi d\phi = \frac{\pi}{2} \sum n A_n^2$$

Thus, the coefficient for the vortex-induced drag

$$C_{Dv} = \pi \cdot AR \sum n A_n^2$$

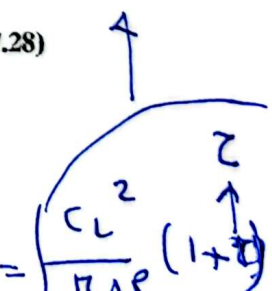
(7.28)

$$\tau = n \sum \left(\frac{A_n}{A_1} \right)^2$$

$$= \pi AR \left[A_1 + \sum_{n=2}^{\infty} n A_n^2 \right]$$

$$\pi AR A_1^2 \left[1 + \sum_{n=2}^{\infty} n \left(\frac{A_n}{A_1} \right)^2 \right] = \left(\frac{C_L^2}{\pi AR} \right) (1 + \tau)$$

Correct



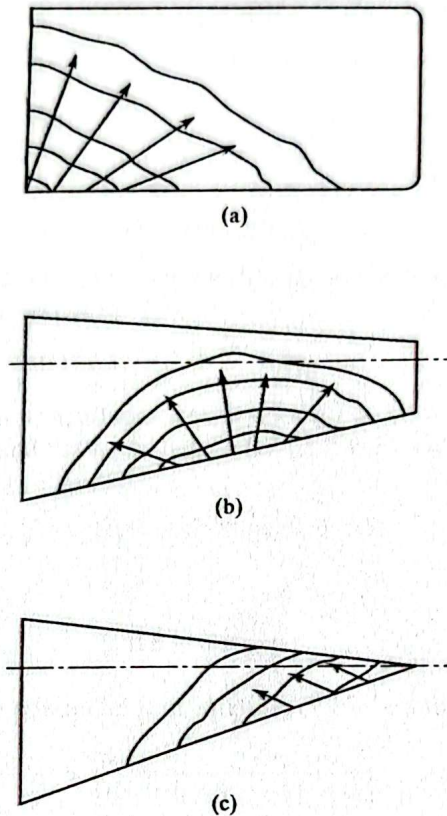


Figure 7.15 Typical stall patterns: (a) rectangular wing, $\lambda = 1.0$; (b) moderately tapered wing, $\lambda = 0.4$; (c) pointed wing, $\lambda = 0.0$.

In order to prevent the stall pattern from beginning in the region of the ailerons, the wing may be given a geometric twist, or washout, to decrease the local angles of attack at the tip (see examples in Table 5.1). The addition of leading-edge slots or slats toward the tip increases the stall angle of attack and is useful in avoiding tip stall and the loss of aileron effectiveness.

EXAMPLE 7.2: Use the monoplane equation to compute the aerodynamic coefficients for a wing

The monoplane equation [i.e., equation (7.26)] will be used to compute the aerodynamic coefficients of a wing for which aerodynamic data are available. The geometry of the wing to be studied is illustrated in Fig. 7.16. The wing, which is unswept at the quarter chord, is composed of NACA 65–210 airfoil sections. Referring to the data of Abbott and von Doenhoff (1949), the zero-lift angle of attack (α_{0l}) for this airfoil is approximately -1.2° across the span. Since the wing is untwisted, the geometric angle of

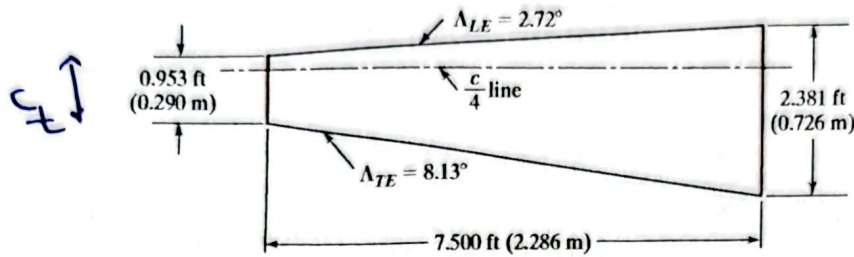


Figure 7.16 Planform for an unswept wing, $AR = 9.00$, $\lambda = 0.40$, airfoil section NACA 65-210.

$$\lambda = \frac{c_t}{c_r}$$

attack is the same at all spanwise positions. The aspect ratio (AR) of the full wing is 9.00, and the taper ratio λ (c_t/c_r) is 0.40. Since the wing planform is trapezoidal,

$$S = \frac{1}{2}(c_r + c_t)b = \frac{1}{2}c_r(1 + \lambda)b$$

and

$$AR = \frac{2b}{c_r + c_t}$$

Therefore, the parameter μ in equation (7.26) becomes:

$$\mu = \frac{ca_0}{4b} = \frac{ca_0}{2(AR) \cdot c_r(1 + \lambda)}$$

where c is the local chord, $c(y)$.

Solution: Since the terms are to be evaluated at spanwise stations for $0 \leq \phi \leq \pi/2$ [i.e., $-s \leq y \leq 0$ (which corresponds to the port wing or left side of the wing)],

If you have
wing with
taper ratio,
use 7.31

$$\mu = \frac{a_0}{2(1 + \lambda)AR} [1 + (\lambda - 1) \cos \phi] = 0.24933(1 - 0.6 \cos \phi) \quad (7.31)$$

where the equivalent lift-curve slope (i.e., for a two-dimensional flow over the airfoil section a_0) has been assumed to be equal to 2π 1/rad. It might be interesting to know that numerical solutions for lift and vortex-drag coefficients were essentially the same for this geometry whether the series representing the spanwise circulation distribution included 4 terms or 10 terms. Therefore, to help the reader perform the required calculations more easily, a four-term series will be used to represent the spanwise loading. Equation (7.26) becomes:

$$\mu(\alpha - \alpha_{0l}) \sin \phi = A_1 \sin \phi(\mu + \sin \phi) + A_3 \sin 3\phi(3\mu + \sin \phi) + A_5 \sin 5\phi(5\mu + \sin \phi) + A_7 \sin 7\phi(7\mu + \sin \phi) \quad (7.32)$$

TABLE 7.1 Values of the Factors for Equation (7.32)

Station	ϕ	$\frac{y}{s}$ (= $\cos \phi$)	$\sin \phi$	$\sin 3\phi$	$\sin 5\phi$	$\sin 7\phi$	μ
1	22.5°	0.92388	0.38268	0.92388	0.92388	0.38268	0.11112
2	45.0°	0.70711	0.70711	0.70711	-0.70711	-0.70711	0.14355
3	67.5°	0.38268	0.92388	-0.38268	-0.38268	0.92388	0.19208
4	90.0°	0.00000	1.00000	-1.00000	1.00000	-1.00000	0.24933

Since there are four coefficients (i.e., A_1 , A_3 , A_5 , and A_7) to be evaluated, equation (7.32) must be evaluated at four spanwise locations. The resultant values for the factors are summarized in Table 7.1. Notice that since we are considering the left side of the wing, the y coordinate is negative.

For a geometric angle of attack of 4° , equation (7.32) becomes:

$$0.00386 = 0.18897A_1 + 0.66154A_3 + 0.86686A_5 + 0.44411A_7$$

for $\phi = 22.5^\circ$ (i.e., $y = -0.92388s$). For the other stations, the equation becomes:

$$0.00921 = 0.60150A_1 + 0.80451A_3 - 1.00752A_5 - 1.21053A_7$$

$$0.01611 = 1.03101A_1 - 0.57407A_3 - 0.72109A_5 + 2.09577A_7$$

$$0.02263 = 1.24933A_1 - 1.74799A_3 + 2.24665A_5 - 2.74531A_7$$

which is four equations in four unknowns. The solution of this system of linear equations yields:

$$\left\{ \begin{array}{l} A_1 = 1.6459 \times 10^{-2} \\ A_3 = 7.3218 \times 10^{-5} \\ A_5 = 8.5787 \times 10^{-4} \\ A_7 = -9.6964 \times 10^{-5} \end{array} \right.$$

Using equation (7.27), the lift coefficient for an angle of attack of 4° is:

$$C_L = A_1 \pi AR = 0.4654$$

The theoretically determined lift coefficients are compared in Fig. 7.17 with data for this wing. In addition to the geometric characteristics already described, the wing has a dihedral angle of 3° , which we are not modeling. The measurements reported by Sivells (1947) were obtained at a Reynolds number of approximately 4.4×10^6 and a Mach number of approximately 0.17, and our theory assumes inviscid, incompressible flow. In spite of these differences, the agreement between the theoretical values and the experimental values is very good.

The spanwise distribution for the local lift coefficient of this wing is presented in Fig. 7.18. As noted by Sivells (1947), the variation of the section

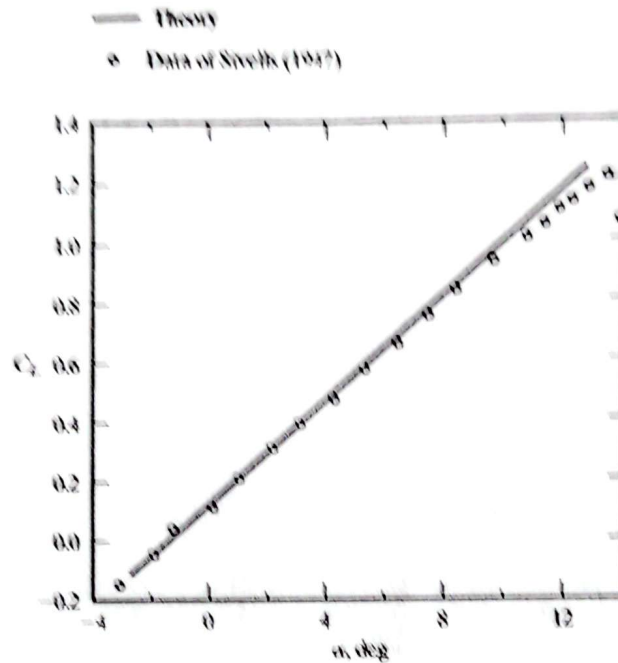


Figure 7.17 Comparison of the theoretical and the experimental lift coefficients for an unswept wing in a subsonic stream (wing is that of Fig. 7.16).

lift coefficient can be used to determine the spanwise position of initial stall. The local lift coefficient is given by:

$$C_l = \frac{\rho_\infty U_\infty \Gamma}{\frac{1}{2} \rho_\infty U_\infty^2 c}$$

where c is the local chord at a given location along the span, $c(y)$. For the trapezoidal wing under consideration is:

$$C_l(\phi) = 2AR(1 + \lambda) \frac{C_l}{c(\phi)} \sum \{A_{2n-1} \sin[(2n-1)\phi]\} \quad (7.33)$$

The theoretical value of the induced drag coefficient for an angle of attack of 4° , as determined using equation (7.29), is:

$$\begin{aligned} C_{De} &= \frac{C_L^2}{\pi AR} \left(1 + \frac{3A_3^2}{A_1^2} + \frac{5A_5^2}{A_1^2} + \frac{7A_7^2}{A_1^2} \right) = \frac{C_L^2}{\pi AR} (1 + \delta) \\ &= 0.00766(1.0136) = 0.00776 \end{aligned}$$

and $\delta = 0.0136$. The theoretically determined induced drag coefficients are compared in Fig. 7.19 with the measured drag coefficients for this wing.

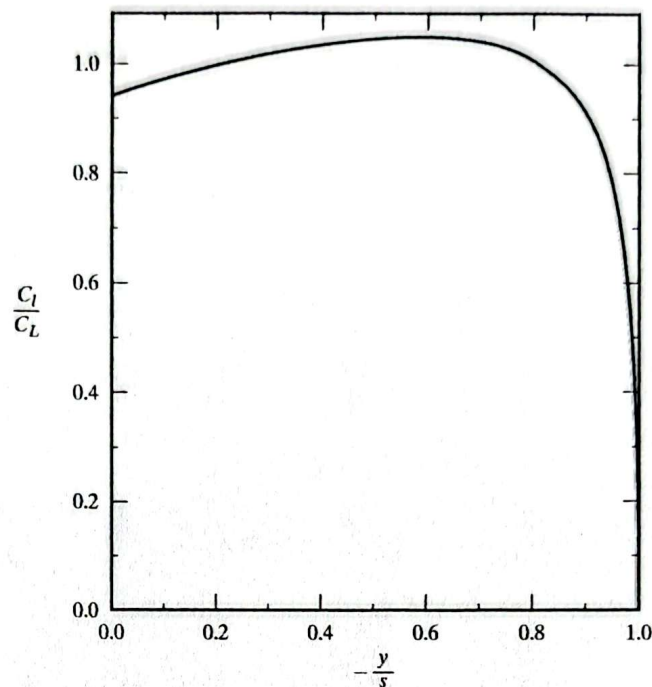


Figure 7.18 Spanwise distribution of the local lift coefficient, $AR = 9$, $\lambda = 0.4$, untwisted wing composed of NACA 65-210 airfoil sections.

As we noted earlier, the theoretical relations developed in this chapter do not include the effects of skin friction. The relatively constant difference between the measured values and the theoretical values is due to the influence of skin friction.

EXAMPLE 7.3: Use the induced drag factor and lift-curve slope parameter to compute the aerodynamic coefficients for a wing

Using the same wing as shown in Example 7.2, in conjunction with the induced drag factor and slope parameter graphs in Fig. 7.14, we can evaluate the effectiveness of a “short-cut” method for estimating wing aerodynamics.

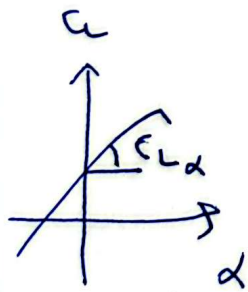
Solution: Equation (7.30) gives the lift-curve slope for a wing in terms of the slope parameter, τ . If we assume that the airfoil lift-curve slope is 2π 1/rad, and find the slope parameter from Fig. 7.14b as $\tau \approx 0.06$, then the wing lift-curve slope is:

$$C_{L_\alpha} = a = \frac{a_0}{1 + \frac{a_0}{\pi AR}(1 + \tau)} = \frac{2\pi \text{ 1/rad}}{1 + \frac{2\pi}{9\pi}(1.06)} = 5.0851/\text{rad} = 0.08881/\text{deg}$$

$$a = C_{L_\alpha \text{ wing}} = \frac{C_{L_\alpha \text{ airfoil}}}{1 + \frac{C_{L_\alpha \text{ airfoil}}}{\pi AR}(1 + \tau)}$$

$\frac{1}{1 + \tau} = e$
wing efficiency factor

$$\tau = n \left(\frac{A_n}{A_1} \right)^2$$



Ex.1) consider a wing with NACA 0012 section, $AR = 7$

$$\begin{cases} a_0 = c_{L\alpha} = 2\pi \\ \alpha_{L_0} = 0 \end{cases}$$

$$AR = \frac{b^2}{S}$$

Use 2-term of Fourier series to find c_L & c_{D_i}

Assume $\lambda = 0.4$, $\alpha = 4^\circ$

taper ratio,

$$\lambda = \frac{c_t}{c_r}$$

Solution:

$$\Gamma = 2bU_\infty (A_1 \sin \phi + A_3 \sin 3\phi)$$

1) compute μ , $\mu = \frac{a_0}{2(1+\lambda)AR} (1 + (\lambda-1)\cos\phi)$ 7.31

$$a_0 = 2\pi$$

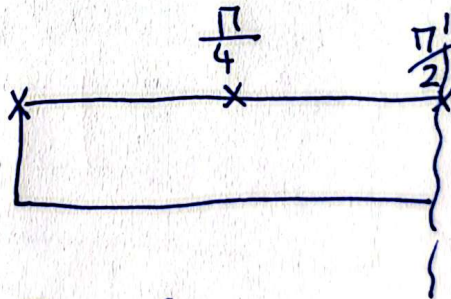
$$AR = 7$$

$$\alpha_{L_0} = 0$$

$$\lambda = 0.4$$

$$\mu = \frac{2\pi}{2(1.4)7} (1 - 0.6 \cos\phi)$$

$$\alpha = 4^\circ$$



consider 2ϕ , $\phi = \frac{\pi}{4}, \frac{\pi}{2}$ ($\phi = 0$ gives $\Gamma = 0$)

① $\phi = \frac{\pi}{4}$, $\mu = 0.18447$

② $\phi = \frac{\pi}{2}$, $\mu = 0.32047$

①

$$\mu (\alpha - \alpha_{L_0}) \sin \phi = A_1 \sin \phi (\mu + \sin \phi) + A_3 \sin 3\phi (3\mu + \sin \phi)$$

One Eq. , two unknowns , write in 2 values of ϕ

first $\phi = 45^\circ \rightarrow 0.18447 \left(\frac{0.06981 - 0}{\text{radian}} \right) \times 0.70717 =$

$$A_1 \times 0.70717 (0.18447 + 0.70717) + A_3 (0.70717) \times$$

$$(3 \times 0.18447 + 0.70717) \Rightarrow \boxed{0.00911 = 0.603045 A_1 + 0.89133 A_3} \quad (1)$$

Second $\phi = 90^\circ$, $\mu = 0.32047$

$$0.32047 (0.06981 - 0) = A_1 (0.32047 + 1) - A_3 (3 \times 0.32047 - 1)$$

$$\boxed{0.02237 = 1.32047 A_1 + 0.038 A_3} \quad (2)$$

Solve (1) and (2) $\Rightarrow \begin{cases} A_1 = 0.016995 \\ A_3 = -0.0018 \end{cases}$

$$C_L = \pi R A_1 = \pi \times 7 \times 0.016995$$

$$C_D = \pi R A_1 (A_1 + 3A_3^2) = \dots$$

(2)

Ex. 2) Consider this Γ , find induced velocity.

$$\Gamma = \Gamma_0 \left(1 - \frac{y^2}{(\frac{b}{2})^2}\right)$$

Solution: Induced velocity is given by:

$$w(y) = \frac{-1}{4\pi} \int_{-\frac{b}{2}}^{\frac{b}{2}} \frac{\frac{d\Gamma}{dy}}{y_0 - y} dy$$

$$\frac{d\Gamma}{dy} = \frac{-8\Gamma_0 y}{b^2} \Rightarrow w(y) = \frac{-2\Gamma_0}{b^2 \pi} \int_{-\frac{b}{2}}^{\frac{b}{2}} \frac{y dy}{y_0 - y}$$

$$= \frac{-2\Gamma_0}{b^2 \pi} \int_{-\frac{b}{2}}^{\frac{b}{2}} \left[\frac{y_0}{y_0 - y} - 1 \right] dy = \frac{2\Gamma_0}{b^2 \pi} \left[b - y \ln \left(\frac{\frac{b}{2} + y_0}{\frac{b}{2} - y_0} \right) \right]$$

Ex. 3) An airplane has wing area of 170 ft^2 , $\underbrace{b}_{\text{span}} = 32 \text{ ft}$

Maximum weight is 2450 lb .

Airfoil of wing has $c_{L_{\alpha_0}} = a_0 = 0.1033 \frac{1}{\text{degree}}$, $\alpha_{L_0} = -3^\circ$

$\tau = 0.12$, velocity = $120 \frac{\text{mi}}{\text{h}}$, standard sea level,

find α and C_{Di} , $e = 0.64$ (consider cruise)

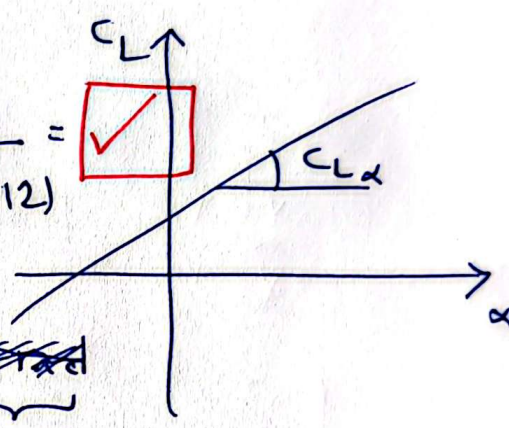
(P-3)

Solution: $AR = \frac{b^2}{S} = \frac{32^2}{170} = 6.02$

$$\left\{ \begin{aligned} C_L &= \frac{L=W}{\frac{1}{2} \rho U_\infty^2 S} = \frac{2450}{\frac{1}{2} \times 0.002377 \times 176^2 \times 170} = 0.39 \quad (1) \\ U &= 120 \text{ mph} \times 3880 \div 3600 = 176 \frac{\text{ft}}{\text{s}} \end{aligned} \right.$$

$$C_{L_\alpha} = a = \frac{a_0}{1 + \frac{a_0}{\pi AR} (1+\epsilon)} = \frac{0.0018}{1 + \frac{0.0018}{\pi \times 6.02} (1.12)} = \boxed{\checkmark}$$

$a_0 = 0.1033 \frac{1}{\text{degree}} = 0.1033 \frac{1}{\frac{\pi}{180}} = \cancel{0.0018}$



Double check!

$$\frac{1 \text{ degree}}{180} \rightarrow \frac{\pi}{180}$$

the other hand $C_L = a(\alpha - \alpha_{L_0}) \quad (2)$

$$= \boxed{\checkmark} (\alpha - (-3)) = \boxed{\checkmark} (\alpha + 3)$$

$D = (2) \Rightarrow \alpha$ can be found!

(P-4)

Part II Compressible Flow

Objective:

The purpose of this part is to provide the student with the fundamental knowledge necessary to understand compressible aerodynamics.

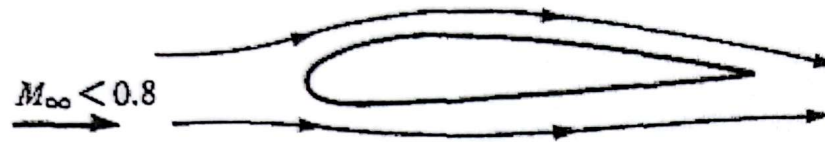
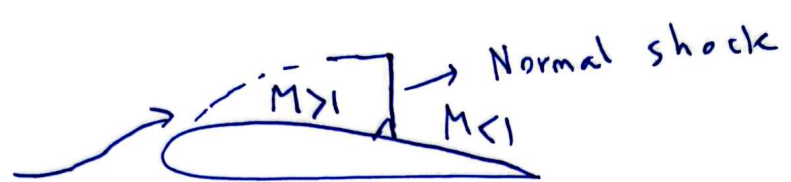
Flow Regimes

$$M = \text{Mach number} = \frac{\text{velocity magnitude}}{\text{speed of sound}} = \frac{|\vec{u}|}{a} = U$$

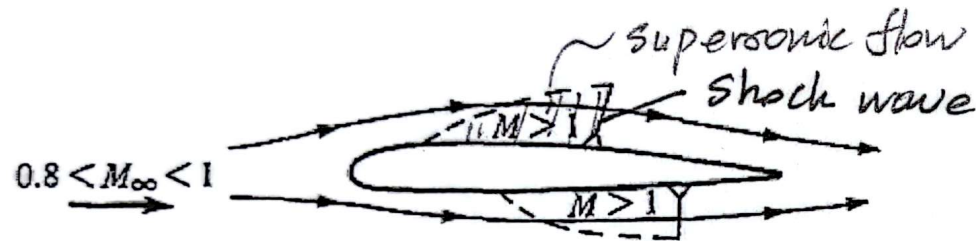
(Incompressible if $M \lesssim 0.3$)

$$M = \sqrt{\gamma R T}$$

(to prove later)

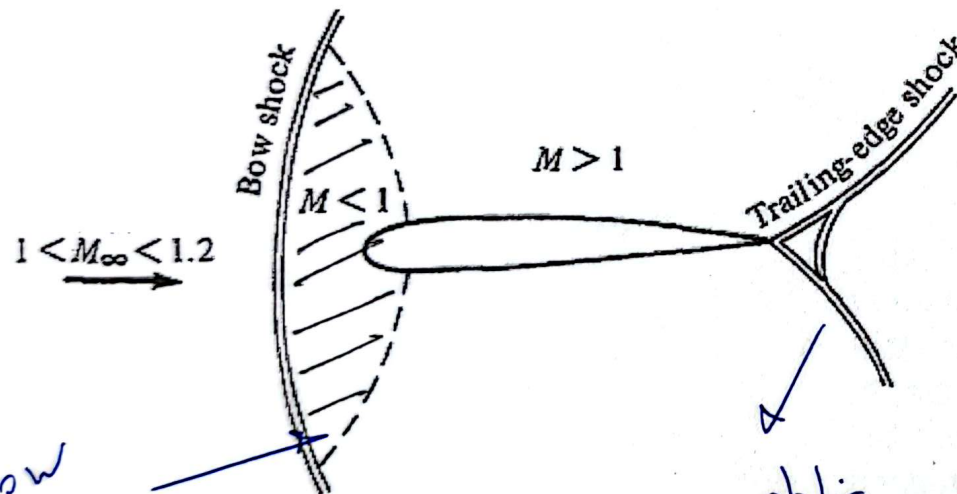


(a) Subsonic flow *everywhere*



(b) Transonic flow
with $M_\infty < 1$

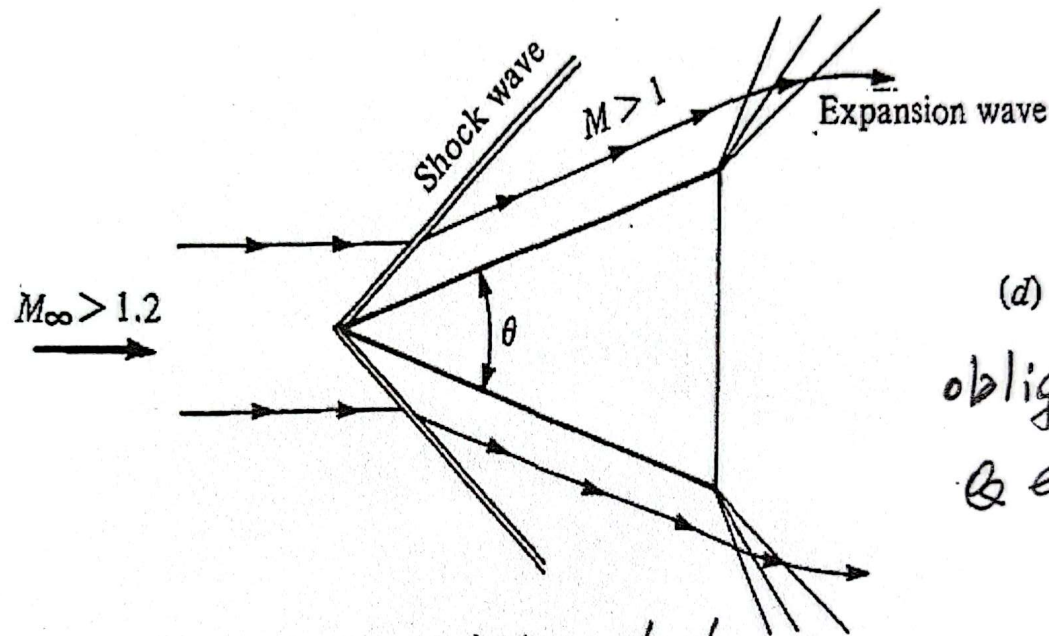
*regions of subsonic
& supersonic flow*



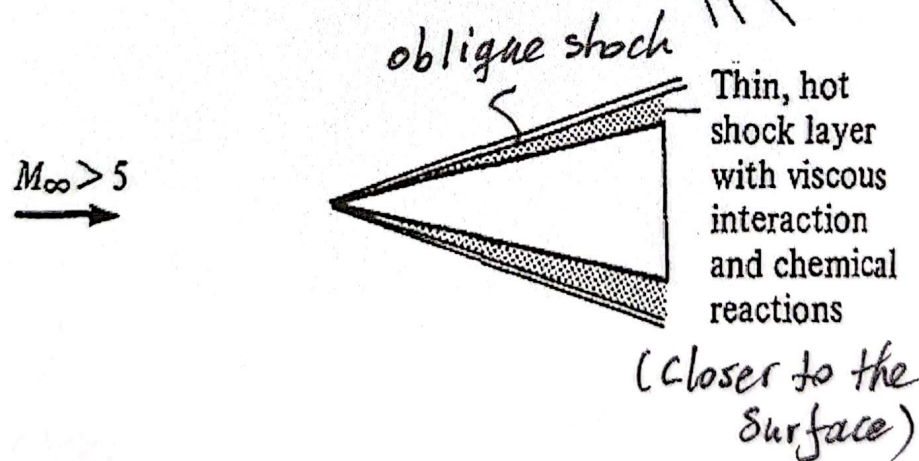
(c) Transonic flow
with $M_\infty > 1$

*bow
shock
or detached
shock*

*oblique
shock*



(d) Supersonic flow
oblique shocks
& expansion waves



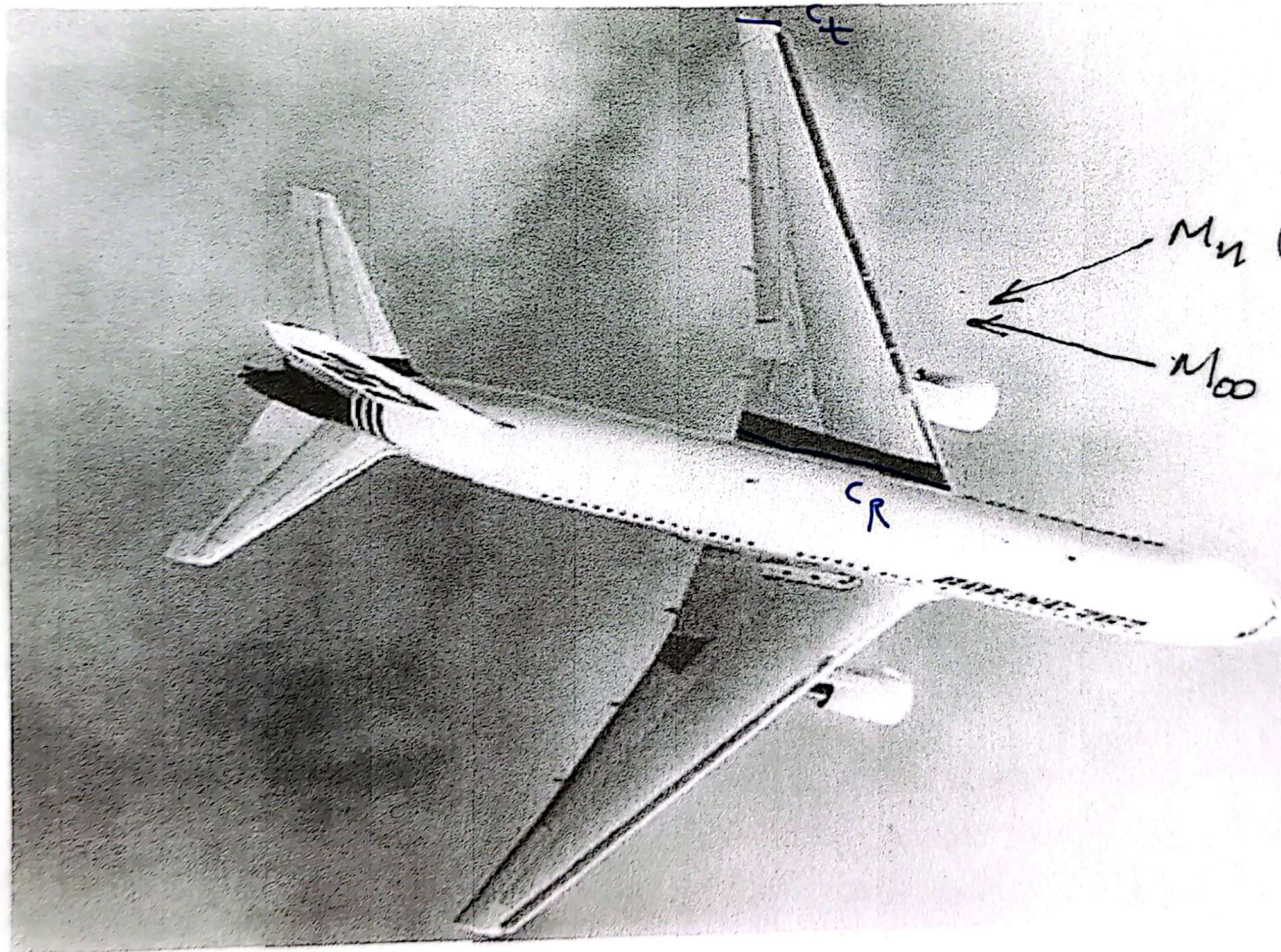
(e) Hypersonic flow
- high temperature near body
possible chemical reactions dissociation
- discontinuous flow possible

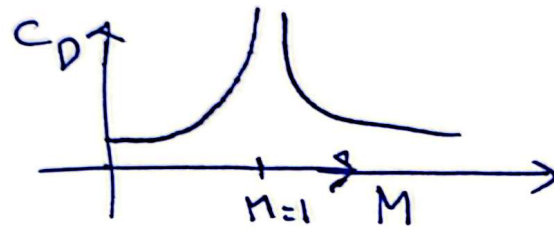
Why do commercial airliners have swept wings?

$$M_n < M_\infty$$

4

→ reduce shock strength
→ reduce shock wave drag

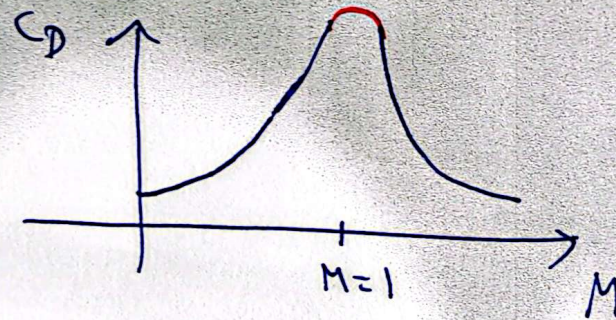




significant⁵
drag
rise

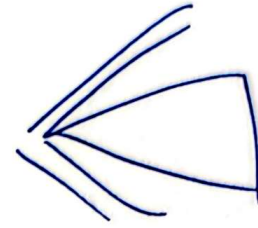
What happens as you pass through the “sound barrier”

transonic

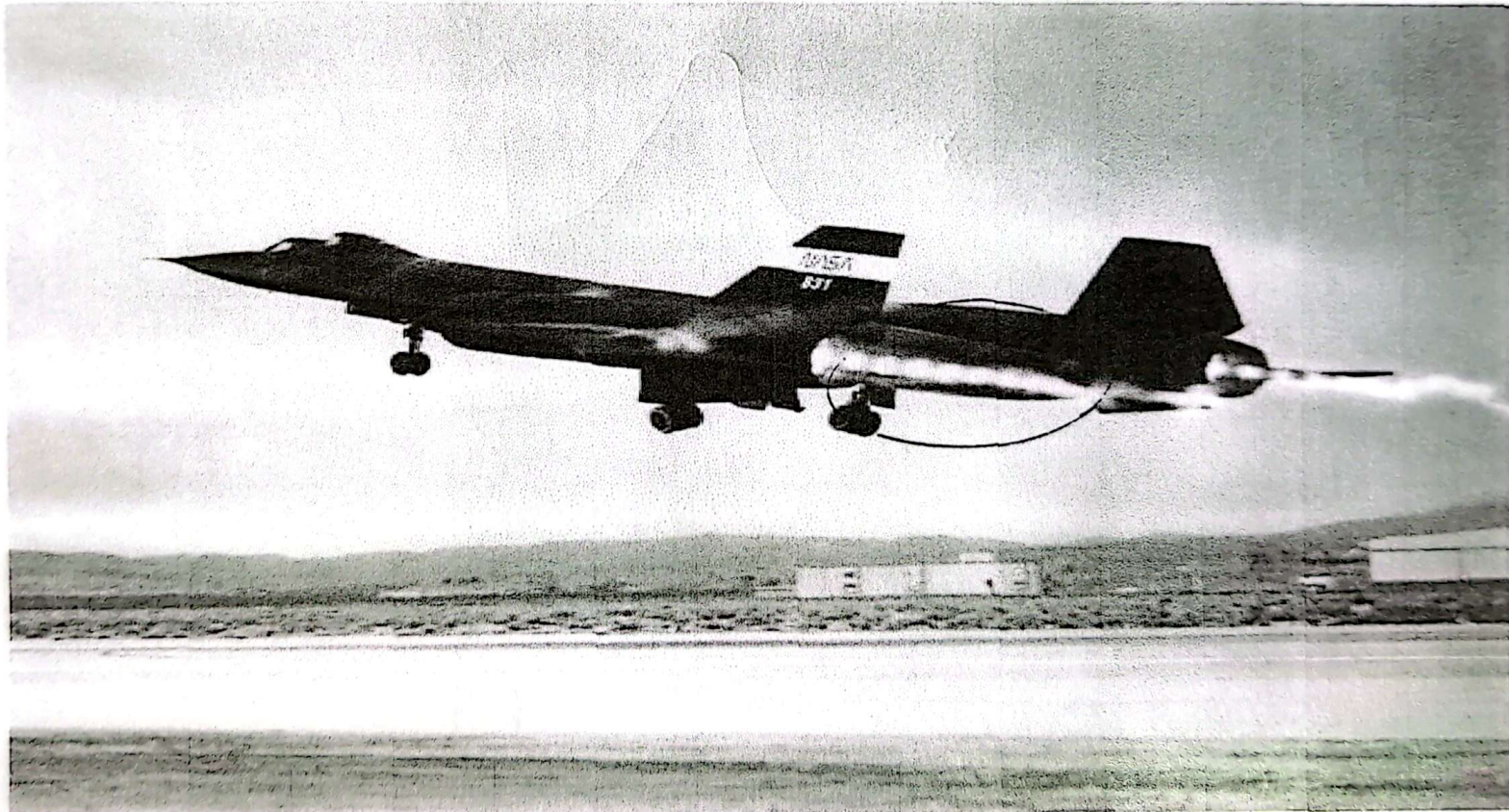


$$D \propto C_D \delta M^2$$





Why do supersonic airfoils have sharp edges? - reduce shock wave drag
 What types of flow can you have with a supersonic nozzle? diamond shapes at the exit



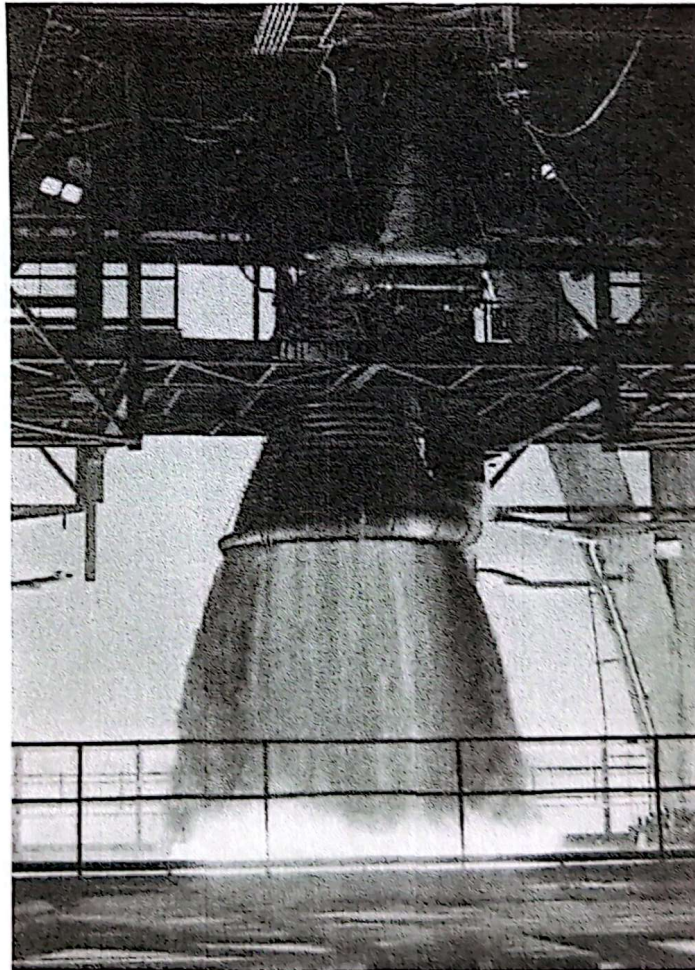
Dryden Flight Research Center EC92-1284 Photographed 1992
 SR-71B take-off with "shock diamonds" in the exhaust. NASA photo



How do you analyze flow in a rocket nozzle?

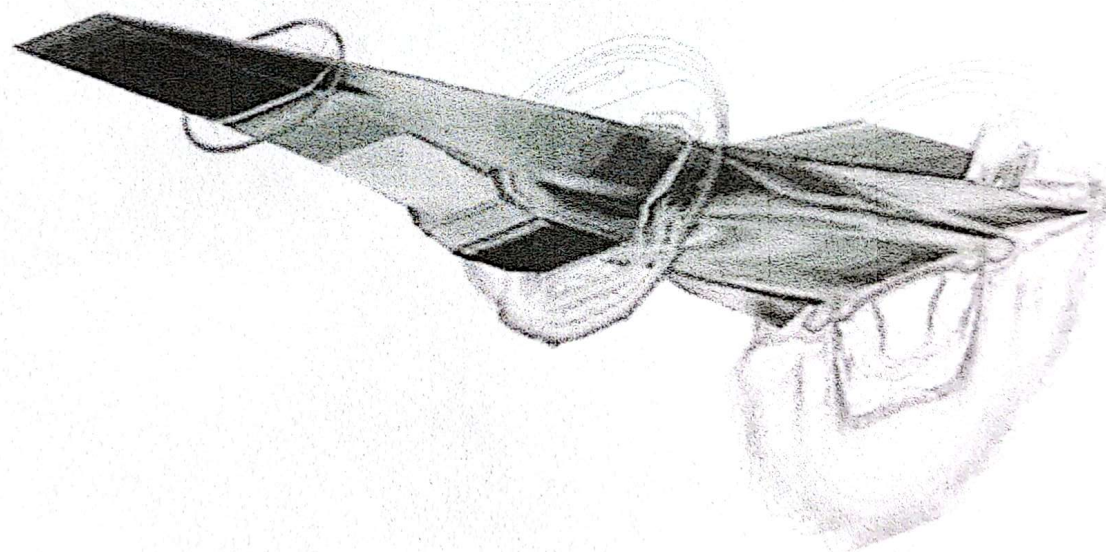


converging/
diverging
nozzle to
achieve $M > 1$



http://www.concordesst.com/pictures/gboag_home.jpg

What happens at hypersonic speeds?



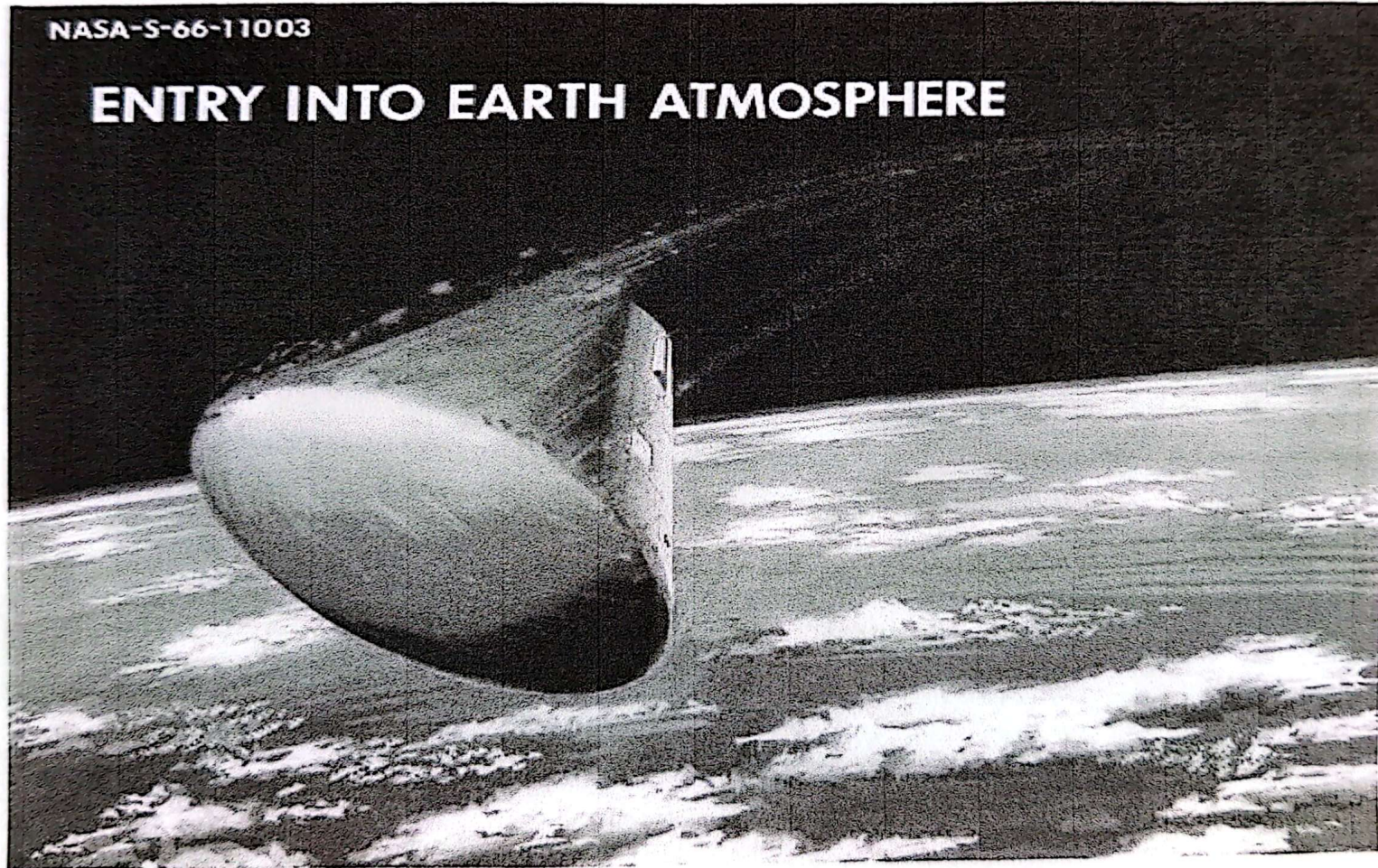
- shock is stronger
- near body
- heat becomes an issue
- wave riders

Dryden Flight Research Center ED97 43968-01
HYPER-X AT MACH 7: This computational fluid dynamic (CFD) image is of the Hyper-X vehicle at the Mach 7 test condition with the engine operating.



What happens during re-entry?

blunt body (heat) $\propto \frac{1}{\sqrt{Re_e}}$



Some Introductory Concepts [8.1-8.3]

Objective: Review of basic thermodynamic concepts (including 2nd law); develop isentropic equations

Brief Review of Thermodynamics [8.1]

p = pressure

ρ = density

T = absolute temperature

$$p = \rho RT$$

$$\left(R = \frac{\bar{R}}{M} \rightarrow \begin{array}{l} \bar{R} \rightarrow \text{universal gas constant (1.10)} \\ M \rightarrow \text{mass of one mole} \end{array} \right. \begin{array}{l} 8314 \text{ J/kg}^\circ\text{K} \\ 28.97 \text{ kg/mole} \end{array} \left. \right)$$

$$R = \text{specific gas constant} = 287 \text{ J/kg}^\circ\text{K} = 1716 \text{ ft lb/(slug }^\circ\text{R)}$$

Or $pv = RT$ v specific volume = $1/\rho$

this is accurate (< 1% error) for low pressures (≤ 1 atm)

moderate temperatures ($\geq 273\text{K}$)

Specific Heats [8.1.1-2]

internal energy: $u_e \equiv u_e(v, T)$ $v = \frac{V}{m}$ (8.1)

enthalpy: $h \equiv h(T, p) = \overbrace{e}^u + pv$ (8.6)

thermally perfect gas: $u_e = u_e(T), \quad h = h(T)$

$$du_e = c_v dT, \quad dh = c_p dT \quad (8.3, 8.10)$$

for c_p, c_v constant $\Rightarrow$ calorically perfect gas

$$u_e = c_v T \quad \text{where:} \quad c_v = \left(\frac{\partial u_e}{\partial T} \right)_v \quad (8.3)$$

$$h = c_p T \quad \text{where:} \quad c_p = \left(\frac{\partial h}{\partial T} \right)_p \quad (8.11)$$

$$c_p - c_v = R \quad \gamma \equiv \frac{c_p}{c_v} \quad (8.15)$$

$$\gamma = \frac{n+2}{n}$$

$n = \#$ of molecular degrees of freedom

$= 1.29$ for monatomic gases $n=3$
 $= 1.4$ for diatomic gases $\approx$ air $n=5$

$= 1.67$ for triatomic gases, eg. CO_2 $n=6$

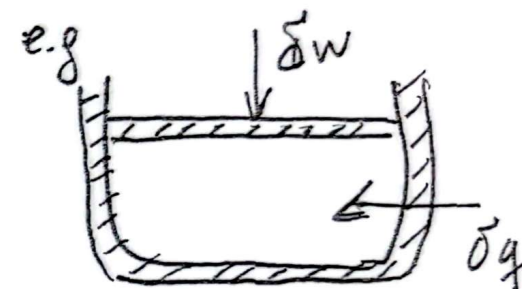
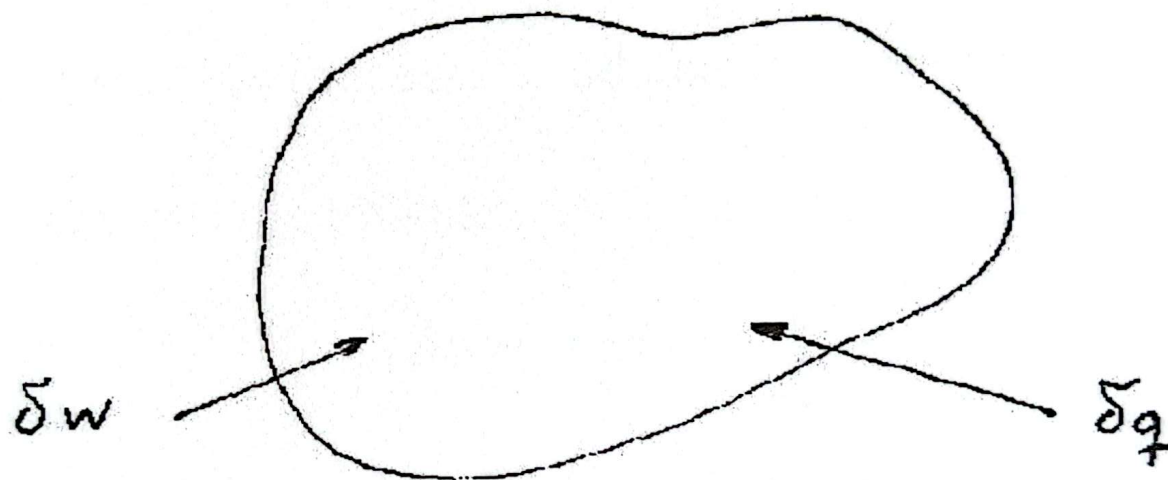
$$\Rightarrow 1 - \frac{c_v}{c_p} = \frac{R}{c_p} \Rightarrow 1 - \frac{1}{\gamma} = \frac{R}{c_p} \Rightarrow$$

$$\Rightarrow c_p = \frac{\gamma R}{\gamma - 1} \quad \text{and} \quad c_v = \frac{R}{\gamma - 1} \quad (8.16)$$

can be found for air (R, γ are known)

First Law of Thermodynamics:

system (fixed mass)



$$\delta q + \delta w = \delta u_e = \text{change in energy}$$

δq = heat added, δw = work done, δu_e = change in energy

- *adiabatic* process: no heat added or taken away
- *reversible* process: no dissipative phenomena occur
- *isentropic* process: adiabatic + reversible

for a reversible process

$$\delta w = -p dv$$

$$S_2 - S_1 = \underbrace{\int \frac{Q}{T}}_{\substack{= 0 \\ \text{if adiabatic}}} + \underbrace{S_{gen}}_{\substack{= 0 \\ \text{if reversible}}}$$

incremental change of volume due to boundary displacement

$$\Rightarrow \delta q - p dv = \delta u_e \quad \Rightarrow \quad S_2 = S_1 \quad \begin{matrix} \text{for} \\ (8.1) \\ \text{isentropic} \end{matrix}$$

(alternative form of 1st law)